MINIMAL EXCLUDED PERMUTATIONS IN A MODEL OF GENOME EVOLUTION

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ABSTRACT. Let Sym_n be the symmetric group on $\{1,\dots,n\}$, equipped with the word metric d_n for the generating set $S_n = \{(a+1\ a+2\ \cdots\ c)^{b-a} : 0 \le a < b < c \le n\}$. In one-line notation, a generator move exchanges two adjacent nonempty segments while preserving the order within each segment. Thus $d_n(\pi, \mathrm{id})$ is the minimum number of such block transpositions needed to sort π, where id denotes the identity of the relevant symmetric group. A standardised restriction is obtained by retaining selected entries of π in their original order and replacing their values by their ranks. For $k \ge 1$, call $\pi \in \mathrm{Sym}_n$ minimal excluded at radius k if $d_n(\pi, \mathrm{id}) > k$, but $d_m(\sigma, \mathrm{id}) \le k$ for every proper standardised restriction $\sigma \in \mathrm{Sym}_m$ of π with $m < n$. These permutations are precisely the minimal forbidden patterns for the class of permutations sortable in at most k block transpositions. We ask how large their degree can be as a function of k.

Let b_k denote the maximum such degree. Cerbai and Ferrari proved $b_k \le 3k+1$. We construct minimal excluded permutations of degree $3k-1$ for every odd $k \ge 3$, and $3k-2$ for every even $k \ge 4$. Each constructed permutation has distance exactly $k+1$, and every standardised one-entry deletion has distance exactly k. Consequently, $3k-1 \le b_k \le 3k+1$ for odd $k \ge 3$ and $3k-2 \le b_k \le 3k+1$ for even $k \ge 4$; in particular, $b_k = 3k + O(1)$ and $b_k/k \to 3$. This determines the optimal leading coefficient in the bound on forbidden-pattern size. In the block-transposition model of genome evolution, entries represent distinct markers and id is their order in a reference chromosome. A minimal excluded permutation rules out every history of at most k transpositions, although omitting any marker makes such a history possible. Our bounds therefore solve the asymptotic extremal problem for the size of a minimal set of markers needed to rule out such evolutionary histories: the maximum is $3k + O(1)$.

1. The problem and the constructions

Let $\Omega_n = \{1,\dots,n\}$, with $\Omega_0 = \emptyset$, and write $\mathrm{Sym}_n = \mathrm{Sym}(\Omega_n)$. Permutations act on the right, so $x\sigma\tau = (x\sigma)\tau$ and $i\pi = \pi_i$.

For $\pi \in \mathrm{Sym}_n$, write $\deg(\pi) = n$; fixed points are included in this count. As traditional in transformation semigroups (and GAP) we will often use words to denote permutations. Therefore, double brackets denote words, $U \parallel V$ denotes concatenation of words, parentheses denote the usual cycle notation, and concatenation of cycles denotes the usual product of permutations. For instance, $\pi = [\![\pi_1,\dots,\pi_n]\!] = \begin{pmatrix} 1 & \cdots & n \\ \pi_1 & \cdots & \pi_n \end{pmatrix}$. For words $U = [\![u_1,\dots,u_r]\!]$ and $V = [\![v_1,\dots,v_s]\!]$, $|U| = r$ and $U \parallel V = [\![u_1,\dots,u_r,v_1,\dots,v_s]\!]$. For a set A, write id_A for its identity map; in particular, $\mathrm{id}_{\Omega_n} = [\![1,\dots,n]\!]$ and $\mathrm{id}_{\Omega_0} = [\![]\!]$.

A *standardised restriction* of $\pi = [\![\pi_1,\dots,\pi_n]\!]$ is obtained by selecting some entries, retaining their order, and replacing the smallest selected value by 1, the next smallest by 2, and so on. It is proper if fewer than n entries are retained.

Formally, for $0 \le m \le n$ and $1 \le i_1 < \cdots < i_m \le n$, write $\sigma = \mathrm{st}([\![\pi_{i_1},\dots,\pi_{i_m}]\!]) \in \mathrm{Sym}_m$, where $\sigma_j = 1 + |\{\ell : 1 \le \ell \le m,\ \pi_{i_\ell} < \pi_{i_j}\}|$ for $1 \le j \le m$. For example, retaining positions $2,4,5$ in the permutation $[\![1,6,2,5,3,4,7,8]\!]$ gives $[\![6,5,3]\!]$, whose standardisation is $\mathrm{st}([\![6,5,3]\!]) = [\![3,2,1]\!]$. The empty selection standardises to id_{Ω_0}.

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For $0 \le a < b < c \le n$, put

$$\begin{aligned} s_{a,b,c} &= [\![1,\dots,a,b+1,\dots,c,a+1,\dots,b,c+1,\dots,n]\!] \\ &= (a+1\ a+2\ \cdots\ c)^{b-a},\ S_n = \{s_{a,b,c} : 0 \le a < b < c \le n\}. \end{aligned} \tag{1}$$

It is easy to check that $s_{a,b,c}^{-1} = s_{a,a+c-b,c}$ and $s_{i-1,i,i+1} = (i\ i+1)$.

Define the left Cayley graph Γ_n by $V(\Gamma_n) = \mathrm{Sym}_n$, $E(\Gamma_n) = \{\{\pi, s\pi\} : \pi \in \mathrm{Sym}_n,\ s \in S_n\}$. The preceding identities give $S_n^{-1} = S_n$ and $\langle S_n \rangle = \mathrm{Sym}_n$, so Γ_n is undirected and connected. Let d_n be its graph distance.

For $\pi = [\![P \mid U \mid V \mid Q]\!]$ with $|P| = a$, $|U| = b-a$, $|V| = c-b$, $|Q| = n-c$, the bars mark the cuts after positions a, b, c, and $s_{a,b,c}[\![P \mid U \mid V \mid Q]\!] = [\![P \mid V \mid U \mid Q]\!]$ (2). The middle words are nonempty; either exterior word may be empty. For example, in Sym_8, $s_{2,4,6} = s_{2,4,6}[\![1,2 \mid 3,4 \mid 5,6 \mid 7,8]\!] = [\![1,2,5,6,3,4,7,8]\!] = (3\,5)(4\,6)$ and $s_{1,2,4}[\![1 \mid 6 \mid 2,5 \mid 3,4,7,8]\!] = [\![1 \mid 2,5 \mid 6 \mid 3,4,7,8]\!]$, $s_{2,4,6}[\![1,2 \mid 5,6 \mid 3,4 \mid 7,8]\!] = [\![1,2 \mid 3,4 \mid 5,6 \mid 7,8]\!] = \mathrm{id}_{\Omega_8}$.

For $k \ge 1$, a permutation $\pi \in \mathrm{Sym}_n$ is *minimal excluded at radius k* if $d_n(\pi, \mathrm{id}_{\Omega_n}) > k$ and $d_m\big(\mathrm{st}([\![\pi_{i_1},\dots,\pi_{i_m}]\!]), \mathrm{id}_{\Omega_m}\big) \le k$ for every $0 \le m < n$ and $1 \le i_1 < \cdots < i_m \le n$. We seek the maximum possible degree n.

Cerbai and Ferrari [2, Theorem 2.5] prove that its degree is at most $3k+1$. Thus, for fixed $k \ge 1$, only degrees $n \le 3k+1$ need be considered. Write b_k for the maximum degree when such permutations exist. Our goal is to provide lower bounds for b_k.

The next lemma contains the first main reduction for the problem.

Lemma 1.1. *Let $n \ge 1$ and $\pi := [\![\pi_1,\dots,\pi_n]\!]$. Every proper standardised restriction of π has distance at most k to its identity if and only if $d_{n-1}\big(\mathrm{st}([\![\pi_1,\dots,\pi_{i-1},\pi_{i+1},\dots,\pi_n]\!]), \mathrm{id}_{\Omega_{n-1}}\big) \le k$ $(1 \le i \le n)$.*

Proof. For $\tau \in \mathrm{Sym}_q$ and $\emptyset \ne A \subseteq \Omega_q$, let $A\tau^{-1} = \{i_1 < \cdots < i_m\}$. Thus

$$\tau\,\mathrm{id}_A = \begin{pmatrix} i_1 & \cdots & i_m \\ \tau_{i_1} & \cdots & \tau_{i_m} \end{pmatrix},\ \mathrm{st}(\tau\,\mathrm{id}_A) = \mathrm{st}([\![\tau_{i_1},\dots,\tau_{i_m}]\!]).$$

In this standardisation, each $x \in A$ is replaced by $|A \cap \Omega_x|$. Take a shortest path $\tau = \tau_0, \tau_1, \dots, \tau_r = \mathrm{id}_{\Omega_q}$, $r = d_q(\tau, \mathrm{id}_{\Omega_q})$, and define, for $0 \le t \le r$, $\eta_t = \mathrm{st}(\tau_t\,\mathrm{id}_A)$. Here A is fixed: $\eta_0 = \mathrm{st}(\tau\,\mathrm{id}_A)$ and $\eta_r = \mathrm{st}(\mathrm{id}_A) = \mathrm{id}_{\Omega_m}$. For $\tau_{t+1} = s_{a,b,c}\tau_t$, put $a' = |A \cap (\Omega_a\tau_t)|$, $b' = |A \cap (\Omega_b\tau_t)|$, $c' = |A \cap (\Omega_c\tau_t)|$. Restricting the four words in (2) to A gives

$$\eta_{t+1} = \begin{cases} s_{a',b',c'}\eta_t, & a' < b' < c', \\ \eta_t, & a' = b' \text{ or } b' = c'. \end{cases}$$

Indeed, the two middle selected words have $b'-a'$ and $c'-b'$ entries; standardisation replaces the same value x by the same rank $|A \cap \Omega_x|$ on both sides. Hence $d_m(\mathrm{st}(\tau\,\mathrm{id}_A), \mathrm{id}_{\Omega_m}) \le \sum_{t=0}^{r-1} d_m(\eta_t, \eta_{t+1}) \le r = d_q(\tau, \mathrm{id}_{\Omega_q})$ (3). The empty selection has distance zero, so this also holds for $m = 0$.

We now prove the two implications.

($\Rightarrow$) Each word $\mathrm{st}([\![\pi_1,\dots,\pi_{i-1},\pi_{i+1},\dots,\pi_n]\!])$ is a proper standardised restriction of π.

($\Leftarrow$) The empty restriction has distance zero. Let $\eta = \mathrm{st}([\![\pi_{i_1},\dots,\pi_{i_m}]\!])$, where $1 \le i_1 < \cdots < i_m \le n$ and $1 \le m < n$. Choose $i \in \Omega_n \setminus \{i_1,\dots,i_m\}$ and put

$$\sigma = \mathrm{st}([\![\pi_1,\dots,\pi_{i-1},\pi_{i+1},\dots,\pi_n]\!]),\ j_\ell = \begin{cases} i_\ell, & i_\ell < i, \\ i_\ell - 1, & i_\ell > i. \end{cases}$$

For $1 \le u, v \le m$, $\sigma_{j_u} < \sigma_{j_v} \iff \pi_{i_u} < \pi_{i_v}$. Thus, for $B = \{\sigma_{j_1},\dots,\sigma_{j_m}\}$, $\eta = \mathrm{st}([\![\sigma_{j_1},\dots,\sigma_{j_m}]\!]) = \mathrm{st}(\sigma\,\mathrm{id}_B)$. Equation (3) and the hypothesis give $d_m(\eta, \mathrm{id}_{\Omega_m}) \le d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) \le k$. □

To prove lower bounds, we need to build some special permutations and prove the needed distance results about them. This is what we do now.

For odd $k = 2m + 1 \geq 3$ and each $j \in \{0, \dots, k-1\}$, define $q_j \in \{0, \dots, k-1\}$ by $q_j \equiv m + j \pmod{k}$. Put $W_j = [\![3j, 3q_j + 2, 3q_j + 1]\!]$ and define $W_0 \parallel \cdots \parallel W_{k-1} = [\![0, a_1, \dots, a_{3k-1}]\!]$, $L_k = [\![a_1, \dots, a_{3k-1}]\!]$ (4). Each value in $\{0, \dots, 3k-1\}$ occurs once, since $j \mapsto q_j$ is a permutation. We include 0 to have $3k$ labels, divided into three sets of size k: $\{0\} \cup \Omega_{3k-1} = \bigcup_{t=0}^{2} \{3j + t : 0 \leq j < k\}$. Each W_j contains one label from each set, and $W_0 = [\![0, 3m + 2, 3m + 1]\!]$. Deleting the initial 0 leaves the labels $1, \dots, 3k-1$. Hence $\{a_1, \dots, a_{3k-1}\} = \{1, \dots, 3k-1\} = \Omega_{3k-1}$ and $L_k = \begin{pmatrix} 1 & \cdots & 3k-1 \\ a_1 & \cdots & a_{3k-1} \end{pmatrix} \in \mathrm{Sym}_{3k-1}$. For even $k \geq 4$, append the two new largest values in decreasing order: $E_k = L_{k-1} \parallel [\![3k-2, 3k-3]\!]$ (5). For $k = 3$, we have $m = 1$. The values of $m + j$, their representatives q_j modulo 3, and the corresponding triples are

j	$m + j$	q_j	W_j
0	1	1	$[\![0, 5, 4]\!]$
1	2	2	$[\![3, 8, 7]\!]$
2	3	0	$[\![6, 2, 1]\!]$

Consequently, $W_0 \parallel W_1 \parallel W_2 = [\![0, 5, 4, 3, 8, 7, 6, 2, 1]\!]$, $L_3 = [\![5, 4, 3, 8, 7, 6, 2, 1]\!]$. For the even parameter $k = 4$, use $L_{k-1} = L_3$ and append $3k - 2 = 10$ and $3k - 3 = 9$, in that order. This gives $E_4 = L_3 \parallel [\![10, 9]\!] = [\![5, 4, 3, 8, 7, 6, 2, 1, 10, 9]\!]$.

Theorem 1.2. *Let $k \geq 3$. Take $\pi = L_k$ and $n = 3k - 1$ when k is odd, and $\pi = E_k$ and $n = 3k - 2$ when k is even. Then $d_n(\pi, \mathrm{id}_{\Omega_n}) = k + 1$ and $d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) = k$, for every standardised one-entry deletion σ of π.*

By Lemma 1.1, these permutations are minimal excluded at radius k. Together with the quoted upper bound, the theorem gives

$$\begin{cases} 3k - 1 \leq b_k \leq 3k + 1, & k \geq 3 \text{ odd}, \\ 3k - 2 \leq b_k \leq 3k + 1, & k \geq 4 \text{ even}, \end{cases} \quad \lim_{k \to \infty} \frac{b_k}{k} = 3.$$

The proof has two parts. For the permutation $\pi \in \mathrm{Sym}_n$ in Theorem 1.2, we prove first that $d_n(\pi, \mathrm{id}_{\Omega_n}) \geq k + 1$, $d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) \leq k$ for every standardised one-entry deletion σ of π. We also prove the inequality $d_n(\pi, \mathrm{id}_{\Omega_n}) \leq d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) + 1$ in Lemma 2.4; together these inequalities give the required equalities.

For even k, we use the result for L_{k-1}.

2. Tools for the proof

2.1. **Restrictions, contraction and direct sums.** For a partial bijection $f : A \to B$ between finite ordered sets of integers, $\mathrm{st}(f)$ is obtained by listing the domain in increasing order and standardising the resulting list of images.

Formally, if $A = \{a_1 < \cdots < a_m\}$, then $\mathrm{st}(f) \in \mathrm{Sym}_m$ is defined by $(\mathrm{st}(f))_i = |\{b \in B : b \leq a_i f\}|$ for $1 \leq i \leq m$. For $A \subseteq \Omega_n$, the product $\pi \,\mathrm{id}_A$ retains the entries of value in A, whereas $\mathrm{id}_A \,\pi$ retains positions in A.

Explicitly, $\pi \,\mathrm{id}_A : A\pi^{-1} \to A$ and $\mathrm{id}_A \,\pi : A \to A\pi$ are both given by $i \mapsto i\pi$ on their respective domains. These are the usual compositions of partial maps [3, § 1.4]; explicitly, $\mathrm{id}_A \,\pi = \pi \,\mathrm{id}_{A\pi}$. Thus deleting the entry of value x gives $\mathrm{st}(\pi \,\mathrm{id}_{\Omega_n \setminus \{x\}})$.

For $\pi, \sigma, \eta \in \mathrm{Sym}_n$, right multiplication by η is an automorphism of Γ_n, since it takes $\{g, sg\}$ to $\{g\eta, s(g\eta)\}$. Hence $d_n(\pi\eta, \sigma\eta) = d_n(\pi, \sigma)$, $d_n(\pi, \sigma) = d_n(\sigma\pi^{-1}, \mathrm{id}_{\Omega_n})$. We shall use the triangle inequality in this graph.

Adjacent entries $[\![j, j + 1]\!]$ form a *successor adjacency*. A *run* of π is a maximal consecutive subword $[\![r, r + 1, \dots, r + t]\!]$, where $t \geq 0$. Retain the first value of every

run and standardise; write $\mathrm{red}(\pi)$ for the result. For example, $\mathrm{red}(\llbracket 4,5,1,2,3,7,8,6\rrbracket) = \llbracket 2,1,4,3\rrbracket$, since the word that we get by retaining the first value of every run is $\llbracket 4,1,7,6\rrbracket$.

Lemma 2.1. *Let $\pi \in \mathrm{Sym}_n$ and let $A \subseteq \Omega_n$ contain at least one value from each run of π. Then $d_n(\pi, \mathrm{id}_{\Omega_n}) = d_{|A|}(\mathrm{st}(\pi\,\mathrm{id}_A), \mathrm{id}_{\Omega_{|A|}})$. Every path from $\mathrm{st}(\pi\,\mathrm{id}_A)$ to its identity lifts to a walk of the same length from π to id_{Ω_n}.*

Proof. For $n = 0$ there is nothing to prove. Let $m = |A| \geq 1$. Split each run immediately before each of its selected entries except the first. List the resulting intervals in increasing value order as $R_j = \llbracket r_j, r_j + 1, \ldots, r_j + h_j - 1\rrbracket$ $(1 \leq j \leq m)$, $h_j \geq 1$. Each R_j contains exactly one value of A. If $\sigma = \mathrm{st}(\pi\,\mathrm{id}_A)$, then $R_1 \parallel \cdots \parallel R_m = \mathrm{id}_{\Omega_n}$, $\pi = R_{\sigma_1} \parallel \cdots \parallel R_{\sigma_m}$. Let $\sigma = \theta_0, \ldots, \theta_r = \mathrm{id}_{\Omega_m}$ be a path. For $0 \leq t \leq r$, define $\pi_t = R_{(\theta_t)_1} \parallel \cdots \parallel R_{(\theta_t)_m}$. Since each R_j contains exactly one value of A and the R_j are listed in increasing value order, $\mathrm{st}(\pi_t\,\mathrm{id}_A) = \theta_t$ for $0 \leq t \leq r$. In particular, $\pi_0 = \pi$ and $\pi_r = \mathrm{id}_{\Omega_n}$. For $0 \leq t < r$, put $H_i = \sum_{j=1}^{i} h_{(\theta_t)_j}$ $(0 \leq i \leq m)$. If $\theta_{t+1} = s_{a,b,c}\theta_t$, then (2) gives $s_{H_a,H_b,H_c}\pi_t = R_{(s_{a,b,c}\theta_t)_1} \parallel \cdots \parallel R_{(s_{a,b,c}\theta_t)_m} = \pi_{t+1}$.

Since each $h_j \geq 1$, $H_b - H_a = \sum_{j=a+1}^{b} h_{(\theta_t)_j} > 0$, $H_c - H_b = \sum_{j=b+1}^{c} h_{(\theta_t)_j} > 0$, and $0 \leq H_a < H_b < H_c \leq H_m = n$. Thus $s_t = s_{H_a,H_b,H_c} \in S_n$ and $\pi_{t+1} = s_t\pi_t$ for $0 \leq t < r$, so $\pi_0, \ldots, \pi_r$ is the required walk. Hence $d_n(\pi, \mathrm{id}_{\Omega_n}) \leq r$. Taking a shortest path and applying (3) proves equality. Figure 1 gives one step with $H_0 = 0$, $H_1 = 2$, $H_2 = 5$. □

Corollary 2.2. *For $\pi \in \mathrm{Sym}_n$ and $m = \deg(\mathrm{red}(\pi))$, $d_n(\pi, \mathrm{id}_{\Omega_n}) = d_m(\mathrm{red}(\pi), \mathrm{id}_{\Omega_m})$.*

Proof. In Lemma 2.1, take A to contain the first value of each run. Then $\mathrm{st}(\pi\,\mathrm{id}_A) = \mathrm{red}(\pi)$. □

For Figure 1, take $\pi = \llbracket 4,5,1,2,3,7,8,6\rrbracket$ and $A = \{1,4,6,7\}$. The vertical maps select the values in A:

$$\pi\,\mathrm{id}_A = \begin{pmatrix} 1 & 3 & 6 & 8 \\ 4 & 1 & 7 & 6 \end{pmatrix}, \qquad \mathrm{st}(\pi\,\mathrm{id}_A) = \llbracket 2,1,4,3\rrbracket,$$

$$(s_{0,2,5}\pi)\,\mathrm{id}_A = \begin{pmatrix} 1 & 4 & 6 & 8 \\ 1 & 4 & 7 & 6 \end{pmatrix}, \quad \mathrm{st}((s_{0,2,5}\pi)\,\mathrm{id}_A) = \llbracket 1,2,4,3\rrbracket.$$

Selecting the positions in A instead gives

$$\mathrm{id}_A\,\pi = \begin{pmatrix} 1 & 4 & 6 & 7 \\ 4 & 2 & 7 & 8 \end{pmatrix}, \ \mathrm{st}(\mathrm{id}_A\,\pi) = \llbracket 2,1,3,4\rrbracket.$$

The entries $1,2,3,4$ represent the intervals $\llbracket 1,2,3\rrbracket$, $\llbracket 4,5\rrbracket$, $\llbracket 6\rrbracket$, $\llbracket 7,8\rrbracket$, respectively. Both vertical maps retain the same values A.

$$\begin{array}{ccc} \llbracket \mid 4,5 \mid 1,2,3 \mid 7,8,6 \rrbracket & \xrightarrow{\quad s_{0,2,5} \quad} & \llbracket \mid 1,2,3 \mid 4,5 \mid 7,8,6 \rrbracket \\ \Big\downarrow \mathrm{st}(\cdot\,\mathrm{id}_A) & & \Big\downarrow \mathrm{st}(\cdot\,\mathrm{id}_A) \\ \llbracket \mid 2 \mid 1 \mid 4,3 \rrbracket & \xrightarrow[\quad s_{0,1,2} \quad]{} & \llbracket \mid 1 \mid 2 \mid 4,3 \rrbracket \end{array}$$

FIGURE 1. Lifting a generator move.

Lemma 2.3. *Let $\pi \in \mathrm{Sym}_n$.*

(i) *Any one entry can be moved to any chosen position in at most one generator step, preserving the relative order of all other entries.*

(ii) *If the values $x, x{+}1$ occur in adjacent positions, in either order, then $\mathrm{st}(\pi\,\mathrm{id}_{\Omega_n\setminus\{x\}}) = \mathrm{st}(\pi\,\mathrm{id}_{\Omega_n\setminus\{x+1\}})$.*

Proof. For (i), write $\pi = [\![P \mid x \mid Q \mid R]\!]$, where P, Q have p, q entries. For $q > 0$, (2) gives $s_{p,p+1,p+q+1}[\![P \mid x \mid Q \mid R]\!] = [\![P \mid Q \mid x \mid R]\!]$. The inverse moves x to the left; for $q = 0$ no step is needed.

For (ii), the two unstandardised deletions are $[\![P, x+1, Q]\!]$ and $[\![P, x, Q]\!]$, in either order. Their middle entries both standardise to x, and every other value y has rank

$$\begin{cases} y, & y < x, \\ y-1, & y > x+1 \end{cases}$$

in both words. Hence $\mathrm{st}([\![P, x+1, Q]\!]) = \mathrm{st}([\![P, x, Q]\!])$. □

Lemma 2.4. *If σ is a standardised one-entry deletion of $\pi \in \mathrm{Sym}_n$, where $n \geq 2$, then $d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) \leq d_n(\pi, \mathrm{id}_{\Omega_n}) \leq d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) + 1$.*

Proof. Let x be the omitted value and $A = \Omega_n \setminus \{x\}$, so $\sigma = \mathrm{st}(\pi\,\mathrm{id}_A)$. Equation (3), applied with $q = n$ and $m = n-1$, gives $d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) \leq d_n(\pi, \mathrm{id}_{\Omega_n})$. By Lemma 2.3(i), there are words P, Q and a permutation π' such that

$$\pi' = \begin{cases} [\![P, x-1, x, Q]\!], & x > 1, \\ [\![P, 1, 2, Q]\!], & x = 1, \end{cases} \quad d_n(\pi, \pi') \leq 1,\ \mathrm{st}(\pi'\,\mathrm{id}_A) = \sigma.$$

The run containing x also contains $x-1$ or 2, respectively; every other run is retained in full. Lemma 2.1 gives $d_n(\pi, \mathrm{id}_{\Omega_n}) \leq d_n(\pi, \pi') + d_n(\pi', \mathrm{id}_{\Omega_n}) \leq 1 + d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}})$. □

Corollary 2.5. *Let $k \geq 1$ and let $\pi \in \mathrm{Sym}_n$ be minimal excluded at radius k. Then $d_n(\pi, \mathrm{id}_{\Omega_n}) = k+1$, and $d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) = k$ for every standardised one-entry deletion σ of π.*

Proof. For every such σ, Lemma 2.4 gives $k+1 \leq d_n(\pi, \mathrm{id}_{\Omega_n}) \leq d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) + 1 \leq k+1$. □

For $\pi \in \mathrm{Sym}_a$ and $\sigma \in \mathrm{Sym}_b$, define their direct sum by $\pi \oplus \sigma = [\![\pi_1, \ldots, \pi_a, a+\sigma_1, \ldots, a+\sigma_b]\!]$. This is the image of (π, σ) in the natural subgroup $\mathrm{Sym}_a \times \mathrm{Sym}_b \leq \mathrm{Sym}_{a+b}$.

Lemma 2.6. *Let $\pi \in \mathrm{Sym}_a$ and $\sigma \in \mathrm{Sym}_b$.*

For $A \subseteq \Omega_a$, $B \subseteq \Omega_b$ and $C = A \cup \{a+j : j \in B\}$, $\mathrm{st}((\pi \oplus \sigma)\,\mathrm{id}_C) = \mathrm{st}(\pi\,\mathrm{id}_A) \oplus \mathrm{st}(\sigma\,\mathrm{id}_B)$.

Then $\max\{d_a(\pi, \mathrm{id}_{\Omega_a}), d_b(\sigma, \mathrm{id}_{\Omega_b})\} \leq d_{a+b}(\pi \oplus \sigma, \mathrm{id}_{\Omega_{a+b}}) \leq d_a(\pi, \mathrm{id}_{\Omega_a}) + d_b(\sigma, \mathrm{id}_{\Omega_b})$.

Proof. Let $P = [\![p_1, \ldots, p_r]\!]$ and $Q = [\![q_1, \ldots, q_t]\!]$ be the image words of $\pi\,\mathrm{id}_A$ and $\sigma\,\mathrm{id}_B$, respectively. Write $\mathrm{st}(P) = [\![u_1, \ldots, u_r]\!]$ and $\mathrm{st}(Q) = [\![v_1, \ldots, v_t]\!]$. Since $p_i \leq a < a + q_j$,

$$\begin{aligned} \mathrm{st}((\pi \oplus \sigma)\,\mathrm{id}_C) &= \mathrm{st}([\![p_1, \ldots, p_r, a+q_1, \ldots, a+q_t]\!]) \\ &= [\![u_1, \ldots, u_r, r+v_1, \ldots, r+v_t]\!] \\ &= \mathrm{st}(\pi\,\mathrm{id}_A) \oplus \mathrm{st}(\sigma\,\mathrm{id}_B). \end{aligned}$$

The same identities hold when $r = 0$ or $t = 0$. In particular, $\mathrm{st}((\pi \oplus \sigma)\,\mathrm{id}_{\Omega_a}) = \pi$, $\mathrm{st}((\pi \oplus \sigma)\,\mathrm{id}_{\{a+1,\ldots,a+b\}}) = \sigma$. Equation (3) gives the lower bounds.

For the upper bound, take shortest paths given by $u_r \cdots u_1 \pi = \mathrm{id}_{\Omega_a}$, $v_t \cdots v_1 \sigma = \mathrm{id}_{\Omega_b}$, $r = d_a(\pi, \mathrm{id}_{\Omega_a})$, $t = d_b(\sigma, \mathrm{id}_{\Omega_b})$, where $u_i \in S_a$ and $v_j \in S_b$.

The embeddings into Sym_{a+b} satisfy

$$\begin{aligned} s_{p,q,h} \oplus \mathrm{id}_{\Omega_b} &= s_{p,q,h} \in S_{a+b} && (0 \leq p < q < h \leq a), \\ \mathrm{id}_{\Omega_a} \oplus s_{p,q,h} &= s_{a+p,a+q,a+h} \in S_{a+b} && (0 \leq p < q < h \leq b). \end{aligned}$$

Therefore $(u_r \oplus \mathrm{id}_{\Omega_b}) \cdots (u_1 \oplus \mathrm{id}_{\Omega_b})(\pi \oplus \sigma) = \mathrm{id}_{\Omega_a} \oplus \sigma$, $(\mathrm{id}_{\Omega_a} \oplus v_t) \cdots (\mathrm{id}_{\Omega_a} \oplus v_1)(\mathrm{id}_{\Omega_a} \oplus \sigma) = \mathrm{id}_{\Omega_{a+b}}$. This gives $d_{a+b}(\pi \oplus \sigma, \mathrm{id}_{\Omega_{a+b}}) \le r + t$, also when one path has no edges. □

Corollary 2.7. *For $\pi \in \mathrm{Sym}_a$ and $r \ge 0$, $d_{a+r}(\pi \oplus \mathrm{id}_{\Omega_r}, \mathrm{id}_{\Omega_{a+r}}) = d_{a+r}(\mathrm{id}_{\Omega_r} \oplus \pi, \mathrm{id}_{\Omega_{a+r}}) = d_a(\pi, \mathrm{id}_{\Omega_a})$.*

Proof. Apply Lemma 2.6 with the other summand equal to id_{Ω_r}, whose distance to the identity is zero. □

For example, $[\![3,1,2]\!] \oplus [\![2,1]\!] = [\![3,1,2,5,4]\!]$. For $\pi \in \mathrm{Sym}_a$, we shall use $d_{a+2}(\pi \oplus [\![2,1]\!], \mathrm{id}_{\Omega_{a+2}}) \le d_a(\pi, \mathrm{id}_{\Omega_a}) + 1$.

2.2. **Cycles and the lower bound.** We write $g^h = h^{-1}gh$ and $[g,h] = g^{-1}h^{-1}gh$.

For $\pi \in \mathrm{Sym}_n$, put $N = n+1$ and $\mathbb{Z}_N = \mathbb{Z}/N\mathbb{Z}$, represented by $0, \ldots, n$. Define

$$\widehat{\pi} \in \mathrm{Sym}(\mathbb{Z}_N),\ x\widehat{\pi} = \begin{cases} 0, & x = 0, \\ x\pi, & 1 \le x \le n. \end{cases}$$

Put $\rho_N = (0\,1\,\cdots\,n)$, $w_\pi = \rho_N^{\widehat{\pi}} = (0\,\pi_1\,\cdots\,\pi_n)$, $\kappa_\pi = \rho_N w_\pi^{-1} = [\rho_N^{-1}, \widehat{\pi}]$ (6). The map $\pi \mapsto w_\pi$ is a bijection onto the set $\mathcal{C}_N$ of N-cycles, and takes id_{Ω_n} to ρ_N. Reading the product defining κ_π gives $y\kappa_\pi = (y+1)w_\pi^{-1}$ $(y \in \mathbb{Z}_N)$ (7), with addition modulo N.

For words $U = [\![u_1, \ldots, u_r]\!]$, $V = [\![v_1, \ldots, v_s]\!]$ and $W = [\![w_1, \ldots, w_t]\!]$, where $r, s, t \ge 0$ and all displayed entries and a, b, c are pairwise distinct, set $(a\,U\,b\,V\,c\,W) := (a\,u_1\,\cdots\,u_r\,b\,v_1\,\cdots\,v_s\,c\,w_1\,\cdots\,w_t)$. In particular, $(a\,U) = (a\,u_1\,\cdots\,u_r)$ and $(a\,[\![]\!]) = (a)$.

Lemma 2.8. *Let $w = (a\,U\,b\,V\,c\,W)$ be an N-cycle, where a, b, c are distinct and U, V, W are possibly empty words. Then $(a\,b\,c)w = (a\,V\,c\,U\,b\,W)$, $(a\,c\,b)w = (a\,W)(b\,U)(c\,V)$.*

Proof. Put $\tau = (a\,b\,c)$, $v = (a\,V\,c\,U\,b\,W)$ and $z = (a\,W)(b\,U)(c\,V)$. Their images are

x	a	b	c
$x(\tau w) = xv$	bw	cw	aw
$x(\tau^{-1}w) = xz$	cw	aw	bw

and, for $x \notin \{a, b, c\}$, $x(\tau w) = xv = xw = x(\tau^{-1}w) = xz$. Thus $\tau w = v$ and $\tau^{-1}w = z$. The permutation v has one cycle and z has three, also when some of U, V, W are empty. □

Corollary 2.9. *Let $w \in \mathcal{C}_N$ and let a, b, c be distinct points of its domain. Exactly one of $(a\,b\,c)w$ and $(a\,c\,b)w$ belongs to $\mathcal{C}_N$.*

Proof. Choose the names b, c so that $w = (a\,U\,b\,V\,c\,W)$. Lemma 2.8 gives one N-cycle and a product of three disjoint nonempty cycles. □

Let Δ_N be the graph with $V(\Delta_N) = \mathcal{C}_N$, $E(\Delta_N) = \{\{w, \tau w\} : w, \tau w \in \mathcal{C}_N,\ \tau \text{ is a 3-cycle on } \mathbb{Z}_N\}$.

Lemma 2.10. *Let $n \ge 0$ and $N = n+1$.*

(i) *For $\pi \in \mathrm{Sym}_n$, put $\pi_0 = 0$. If $0 \le a < b < c \le n$, $\sigma = s_{a,b,c}\pi$ and $\tau = (\pi_a\,\pi_b\,\pi_c)$, then $w_\sigma = \tau w_\pi$, $\kappa_\sigma = \kappa_\pi \tau^{-1}$ (8).*

(ii) *The map $\pi \mapsto w_\pi$ is a graph isomorphism $\Gamma_n \longrightarrow \Delta_N$.*

Proof. For (i), put $P = [\![\pi_1, \ldots, \pi_a]\!]$, $U = [\![\pi_{a+1}, \ldots, \pi_b]\!]$, $V = [\![\pi_{b+1}, \ldots, \pi_c]\!]$, $Q = [\![\pi_{c+1}, \ldots, \pi_n]\!]$. Then $\pi = [\![P \mid U \mid V \mid Q]\!]$ and $\sigma = [\![P \mid V \mid U \mid Q]\!]$. Lemma 2.8 gives $w_\sigma = (0\,P\,V\,U\,Q) = (\pi_a\,\pi_b\,\pi_c)(0\,P\,U\,V\,Q) = \tau w_\pi$, and hence $\kappa_\sigma = \rho_N(\tau w_\pi)^{-1} = \rho_N w_\pi^{-1}\tau^{-1} = \kappa_\pi \tau^{-1}$.

For (ii), the inverse bijection is $(0\,x_1\,\cdots\,x_n) \mapsto [\![x_1, \ldots, x_n]\!]$. Part (i) maps every edge of Γ_n to an edge of Δ_N. Conversely, suppose $\{w_\pi, \tau w_\pi\} \in E(\Delta_N)$ and write

$\{i \in \{0,\dots,n\} : \pi_i\tau \neq \pi_i\} = \{a,b,c\}$, $a<b<c$. Then $\tau \in \{(\pi_a\,\pi_b\,\pi_c),(\pi_a\,\pi_c\,\pi_b)\}$. By Lemma 2.8, the condition $\tau w_\pi \in \mathcal{C}_N$ forces $\tau = (\pi_a\,\pi_b\,\pi_c)$. Part (i) now gives $\tau w_\pi = w_{s_{a,b,c}\pi}$, as required. □

For example, take the first move from Section 1: $\pi = [\![1,6,2,5,3,4,7,8]\!]$, $(a,b,c) = (1,2,4)$, $\sigma = s_{1,2,4}\pi = [\![1,2,5,6,3,4,7,8]\!]$, $\tau = (\pi_1\,\pi_2\,\pi_4) = (1\,6\,5)$. Thus $\tau w_\pi = (1\,6\,5)(0\,1\,6\,2\,5\,3\,4\,7\,8) = (0\,1\,2\,5\,6\,3\,4\,7\,8) = w_\sigma$, $\tau^{-1}w_\pi = (0\,1\,3\,4\,7\,8)(2\,5)(6) \notin \mathcal{C}_9$.

For a permutation γ of a finite set, write $\gamma = \gamma_1\cdots\gamma_t$ for its disjoint cycle decomposition, including each fixed point as a 1-cycle. Define $c_+(\gamma) = |\{i : \mathrm{sgn}(\gamma_i) = 1\}|$.

If γ_i has length ℓ_i, then $\mathrm{sgn}(\gamma_i) = (-1)^{\ell_i-1}$, $c_+(\gamma) = |\{i : \ell_i \equiv 1 \pmod 2\}|$.

Lemma 2.11. *Let γ be a permutation and τ a 3-cycle on its domain. Then $c_+(\gamma\tau^{-1}) - c_+(\gamma) \in \{-2,0,2\}$.*

Suppose $\gamma = \gamma_1\cdots\gamma_t$ with $\mathrm{sgn}(\gamma_i) = 1$ for every i. Then $c_+(\gamma\tau^{-1}) - c_+(\gamma) = 2$ if and only if some cycle factor is $\eta = (a\,A\,b\,B\,c\,C)$ with $|A|,|B|,|C| \in 2\mathbb{Z}$ and $\tau = (a\,b\,c)$. In that case $\eta\tau^{-1} = (a\,A)(b\,B)(c\,C)$, and all other factors are unchanged.

Proof. For $\tau = (a\,b\,c)$ and every point x in the domain of γ,

$$x(\gamma\tau^{-1}) = \begin{cases} c, & x\gamma = a,\\ a, & x\gamma = b,\\ b, & x\gamma = c,\\ x\gamma, & x\gamma \notin \{a,b,c\}. \end{cases}$$

When the three selected points belong to distinct cycle factors, the two orientations give $(a\,A)(b\,B)(c\,D)(a\,b\,c)^{-1} = (a\,A\,c\,D\,b\,B)$, $(a\,A)(b\,B)(c\,D)(a\,c\,b)^{-1} = (a\,A\,b\,B\,c\,D)$. When they meet two factors, write $\eta = (a\,A\,b\,B)$ and $\zeta = (c\,D)$. Then $\eta\zeta(a\,b\,c)^{-1} = (a\,A)(b\,B\,c\,D)$, $\eta\zeta(a\,c\,b)^{-1} = (b\,B)(a\,A\,c\,D)$. When all three belong to $\eta = (a\,A\,b\,B\,c\,C)$, we have $\eta(a\,b\,c)^{-1} = (a\,A)(b\,B)(c\,C)$, $\eta(a\,c\,b)^{-1} = (a\,A\,c\,C\,b\,B)$ (9).

Let r,s be the numbers of affected factors before and after multiplication, and let o,o' be their numbers of odd lengths. These products give $(r,s) \in \{(3,1),(2,2),(1,3),(1,1)\}$. The sum M of the affected lengths is unchanged. Therefore $o \equiv M \equiv o' \pmod 2$, $0 \le o \le r$, $0 \le o' \le s$, which gives $o' - o \in \{-2,0,2\}$ in each of the four cases.

Suppose every original factor is even, so $o = r$. We prove the two implications in the final assertion.

($\Rightarrow$) If $o' - o = 2$, then $o' = r+2 \le s$. The only possibility is $(r,s) = (1,3)$ and $o' = 3$. By (9), $\eta = (a\,A\,b\,B\,c\,C)$, $\tau = (a\,b\,c)$, $\eta\tau^{-1} = (a\,A)(b\,B)(c\,C)$. All three resulting cycles have odd length; hence each of A,B,C has an even number of entries.

($\Leftarrow$) Under these conditions, $c_+(\eta) = 1$ and $c_+(\eta\tau^{-1}) = 3$. All other factors are unchanged, so $c_+(\gamma\tau^{-1}) - c_+(\gamma) = 3 - 1 = 2$. □

For $\pi \in \mathrm{Sym}_n$, put $\nu(\pi) = \frac{n+1-c_+(\kappa_\pi)}{2}$ (10).

If the cycle lengths of κ_π are $\ell_1,\dots,\ell_t$, then $n+1 = \sum_{i=1}^t \ell_i \equiv |\{i : \ell_i \text{ is odd}\}| = c_+(\kappa_\pi) \pmod 2$. Since $0 \le c_+(\kappa_\pi) \le n+1$, we have $\nu(\pi) \in \mathbb{Z}_{\ge 0}$. Also $\nu(\mathrm{id}_{\Omega_n}) = 0$.

Lemma 2.12. *For every $\pi \in \mathrm{Sym}_n$, $d_n(\pi,\mathrm{id}_{\Omega_n}) \ge \nu(\pi)$. A generator step changes ν by an element of $\{-1,0,1\}$.*

Proof. For $\sigma = s_{a,b,c}\pi$, Equation (8) and Lemma 2.11 give $\nu(\sigma) - \nu(\pi) = \frac{c_+(\kappa_\pi) - c_+(\kappa_\pi\tau^{-1})}{2} \in \{-1,0,1\}$, $\tau = (\pi_a\,\pi_b\,\pi_c)$. For a shortest path $\pi = \pi_0,\dots,\pi_r = \mathrm{id}_{\Omega_n}$, $\nu(\pi) = \sum_{i=0}^{r-1}(\nu(\pi_i) - \nu(\pi_{i+1})) \le r = d_n(\pi,\mathrm{id}_{\Omega_n})$. □

Lemma 2.13. *Let $\pi \in \mathrm{Sym}_n$. Suppose that every cycle factor of κ_π has length 1, 3 or 5.*

Let τ be a 3-cycle with $\tau w_\pi \in \mathcal{C}_{n+1}$, and define σ by $w_\sigma = \tau w_\pi$. Then $\nu(\sigma) = \nu(\pi) - 1$ if and only if τ is a 3-cycle factor of κ_π, or some 5-cycle factor is $(a\,b\,c\,d\,e)$ and $\tau \in \{(a\,b\,c),(b\,c\,d),(c\,d\,e),(d\,e\,a),(e\,a\,b)\}$.

Proof. By Lemma 2.11, $\nu(s\pi) = \nu(\pi) - 1$ requires one factor to split into three odd lengths. For lengths $1, 3, 5$, the possible triples of lengths are, up to order, $3 = 1+1+1$, $5 = 1+1+3$. The corresponding products, up to cyclic rotation of the factor, are $(a\,b\,c)(a\,b\,c)^{-1} = (a)(b)(c)$, $(a\,b\,c\,d\,e)(a\,b\,c)^{-1} = (a)(b)(c\,d\,e)$. These products increase c_+ by two. Lemma 2.10 gives the remaining condition $\tau w_\pi \in \mathcal{C}_{n+1}$. □

Lemma 2.14. *Let $\pi \in \mathrm{Sym}_n$ and $q = \nu(\pi) \geq 1$. Suppose that $d_n(s\pi, \mathrm{id}_{\Omega_n}) \geq q$ for every $s \in S_n$ satisfying $\nu(s\pi) = q - 1$. Then $d_n(\pi, \mathrm{id}_{\Omega_n}) \geq q + 1$.*

Proof. Suppose $d_n(\pi, \mathrm{id}_{\Omega_n}) \leq q$. Lemma 2.12 gives a shortest path $\pi = \pi_0, \pi_1, \ldots, \pi_q = \mathrm{id}_{\Omega_n}$. Then $q = \nu(\pi) = \sum_{i=0}^{q-1}(\nu(\pi_i) - \nu(\pi_{i+1}))$, $\nu(\pi_i) - \nu(\pi_{i+1}) \leq 1$. There are q summands, so each equals 1. In particular, $\pi_1 = s\pi$ for some $s \in S_n$ and $\nu(\pi_1) = q - 1$. The hypothesis and the remaining path give $q \leq d_n(\pi_1, \mathrm{id}_{\Omega_n}) \leq q-1$, a contradiction. □

3. Proof of Theorem 1.2

3.1. **The odd case.** Fix $k = 2m+1 \geq 3$ and put $N = 3k = 6m+3$. The words W_j and the permutation L_k are defined in (4).

Read labels modulo N and subscripts modulo k. Since $x\rho_N^3 = x + 3 \pmod N$ and $q_{j+1} \equiv q_j + 1 \pmod k$, adding 3 to each entry gives $[\![3j, 3q_j + 2, 3q_j + 1]\!] \longmapsto [\![3(j+1), 3q_{j+1} + 2, 3q_{j+1} + 1]\!] = W_{j+1}$. Thus $w_{L_k}^{\rho_N^3} = w_{L_k}$. The commutator factors are shown in Figure 2. The labels on the circumference occur in the cyclic order of w_{L_5}, and the arrows are the five factors of κ_{L_5}.

For each arrowed factor $(a\,b\,c)$, there are integers $0 < i < j < 15$ such that $a(w_{L_5})^i = c$ and $a(w_{L_5})^j = b$.

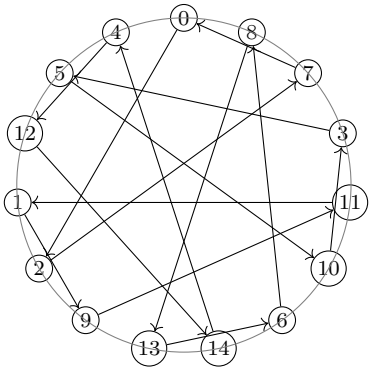


FIGURE 2. The permutation L_5.

Lemma 3.1. *The commutator of L_k is*

$$\begin{aligned}\kappa_{L_k} = &\prod_{s=0}^{m}(3s,\ 3s+2,\ 3m+3s+1)\\ &\cdot\prod_{s=0}^{m-1}(3s+1,\ 3m+3s+3,\ 3m+3s+5).\end{aligned} \tag{11}$$

The displayed cycles are disjoint and cover $\mathbb{Z}_N$.

Moreover, for each displayed factor $(a\,b\,c)$, there exist integers $0 < \ell_1 < \ell_2 < N$ such that $a(w_{L_k})^{\ell_1} = c$, $a(w_{L_k})^{\ell_2} = b$.

Proof. Let $0 \leq s \leq m$. Applying the predecessor identity (7) to ρ_N and w_{L_k}, with addition modulo N, gives $3s \mapsto 3s+2$, $3s+2 \mapsto 3m+3s+1$, $3m+3s+1 \mapsto 3s$, and hence gives the first product in (11). Let $0 \leq s < m$. The same calculation gives

$3s+1 \mapsto 3m+3s+3$, $3m+3s+3 \mapsto 3m+3s+5$, $3m+3s+5 \mapsto 3s+1$, and hence gives the second product.

Writing $\sqcup$ for disjoint union, the six sets of labels in (11) satisfy

$$\{3s : 0 \le s \le m\} \sqcup \{3m+3s+3 : 0 \le s < m\} = \{3j : 0 \le j \le 2m\},$$
$$\{3m+3s+1 : 0 \le s \le m\} \sqcup \{3s+1 : 0 \le s < m\} = \{3j+1 : 0 \le j \le 2m\},$$
$$\{3s+2 : 0 \le s \le m\} \sqcup \{3m+3s+5 : 0 \le s < m\} = \{3j+2 : 0 \le j \le 2m\}.$$

The three sets on the right partition $\mathbb{Z}_N = \{0, \dots, 6m+2\}$, so every label occurs in exactly one displayed cycle.

In w_{L_k}, the entries of W_j occupy positions $3j, 3j+1, 3j+2$ in the word starting at 0. The definition of q_j gives

a	$a(w_{L_k})^{\ell_1}$	$a(w_{L_k})^{\ell_2}$	(ℓ_1, ℓ_2)
$3s$	$3m+3s+1$	$3s+2$	$(2, 3m+4)$ $(0 \le s \le m)$
$3s+1$	$3m+3s+5$	$3m+3s+3$	$(3m+2, 6m+1)$ $(0 \le s < m)$

In each row, $0 < \ell_1 < \ell_2 < N$, and the factor in (11) is $(a,\, a(w_{L_k})^{\ell_2},\, a(w_{L_k})^{\ell_1})$. This proves the reversed cyclic order. □

Lemma 3.2. *For every odd $k \ge 3$, $d_{3k-1}(L_k, \mathrm{id}_{\Omega_{3k-1}}) \ge k+1$.*

Proof. Lemma 3.1 gives k disjoint 3-cycles on $3k$ labels, so $\nu(L_k) = (3k-k)/2 = k$.

For every 3-cycle factor τ of κ_{L_k}, Lemma 3.1 and Lemma 2.8 give $\tau w_{L_k} \notin \mathcal{C}_{3k}$. Hence Lemma 2.13 gives $\nu(sL_k) \ge \nu(L_k) = k$ $(s \in S_{3k-1})$. Lemma 2.14 therefore gives $d_{3k-1}(L_k, \mathrm{id}_{\Omega_{3k-1}}) \ge k+1$. □

Lemma 3.3. *Let $N \ge 3$, $\pi \in \mathrm{Sym}_{N-1}$ and $t \in \mathbb{Z}_N$, and define π_t by $w_{\pi_t} = w_\pi^{\rho_N^t}$. Then $d_{N-1}(\pi_t, \mathrm{id}_{\Omega_{N-1}}) = d_{N-1}(\pi, \mathrm{id}_{\Omega_{N-1}})$.*

For nonzero $x, y \in \mathbb{Z}_N$ with $y = x+t$, we have $d_{N-2}(\mathrm{st}(\pi\, \mathrm{id}_{\Omega_{N-1}\setminus\{x\}}), \mathrm{id}_{\Omega_{N-2}}) = d_{N-2}(\mathrm{st}(\pi_t\, \mathrm{id}_{\Omega_{N-1}\setminus\{y\}}), \mathrm{id}_{\Omega_{N-2}})$.

Proof. For $g = \rho_N^t$, $\{w, \tau w\} \in E(\Delta_N) \Longrightarrow \{w^g, \tau^g w^g\} \in E(\Delta_N)$, $\rho_N^g = \rho_N$. The inverse map is conjugation by g^{-1}, so this is an automorphism fixing ρ_N. Lemma 2.10 gives the first assertion; for $N = 1, 2$ the graphs have one vertex.

Put $A = \mathbb{Z}_N \setminus \{x\}$ and $B = \mathbb{Z}_N \setminus \{y\}$.

Let $i = x\pi^{-1}$ and $j = y\pi_t^{-1}$. Define cycles on A, B by $\rho_z = (0\, 1 \cdots z-1\, z+1 \cdots N-1)$ $(z \in \{x, y\})$, $w_x = (0\, \pi_1 \cdots \pi_{i-1}\, \pi_{i+1} \cdots \pi_{N-1})$, $w_y = (0\, (\pi_t)_1 \cdots (\pi_t)_{j-1}\, (\pi_t)_{j+1} \cdots (\pi_t)_{N-1})$. Let $h : A \to B$ be given by $zh = z+t \pmod N$.

Conjugation relabels the surviving cyclic orders, so $h^{-1}\rho_x h = \rho_y$, $h^{-1} w_x h = w_y$.

For $z \in \{x, y\}$, the increasing bijection $s_z : \mathbb{Z}_N \setminus \{z\} \to \mathbb{Z}_{N-1}$ is

$$us_z = \begin{cases} u, & u < z, \\ u-1, & u > z. \end{cases}$$

Writing $\sigma = \mathrm{st}(\pi\, \mathrm{id}_{\Omega_{N-1}\setminus\{x\}})$ and $\tau = \mathrm{st}(\pi_t\, \mathrm{id}_{\Omega_{N-1}\setminus\{y\}})$, we have $s_x^{-1}\rho_x s_x = s_y^{-1}\rho_y s_y = \rho_{N-1}$, $s_x^{-1} w_x s_x = w_\sigma$, $s_y^{-1} w_y s_y = w_\tau$.

Put $\zeta = s_x^{-1} h s_y$. These identities give $\zeta^{-1}\rho_{N-1}\zeta = \rho_{N-1}$, $\zeta^{-1} w_\sigma \zeta = w_\tau$. If $q = 0\zeta$, then, for $i \in \mathbb{Z}_{N-1}$, $i\zeta = (0\rho_{N-1}^i)\zeta = (0\zeta)\rho_{N-1}^i = q+i$, so $\zeta = \rho_{N-1}^q$. The first assertion, in degree $N-2$, now gives $d_{N-2}(\sigma, \mathrm{id}_{\Omega_{N-2}}) = d_{N-2}(\tau, \mathrm{id}_{\Omega_{N-2}})$. □

Lemma 3.4. *Every standardised one-entry deletion of L_{2m+1} has the same distance to the identity as one of $A_m = \mathrm{st}(L_{2m+1}\, \mathrm{id}_{\Omega_{6m+2}\setminus\{1\}})$, $B_m = \mathrm{st}(L_{2m+1}\, \mathrm{id}_{\Omega_{6m+2}\setminus\{3\}})$.*

Proof. Let $x \in \Omega_{N-1}$ and put

$$y = \begin{cases} 1, & x \equiv 1 \pmod 3, \\ 2, & x \equiv 2 \pmod 3, \\ 3, & x \equiv 0 \pmod 3. \end{cases}$$

Since $y - x$ is a multiple of 3 and $w_{L_k}^{\rho_N^3} = w_{L_k}$, Lemma 3.3 gives $d_{N-2}(\mathrm{st}(L_k \,\mathrm{id}_{\Omega_{N-1}\setminus\{x\}}), \mathrm{id}_{\Omega_{N-2}}) = d_{N-2}(\mathrm{st}(L_k \,\mathrm{id}_{\Omega_{N-1}\setminus\{y\}}), \mathrm{id}_{\Omega_{N-2}})$. Both x and y are nonzero. Moreover, $W_{m+1} = [\![3m+3, 2, 1]\!]$, so Lemma 2.3(ii) gives $\mathrm{st}(L_k \,\mathrm{id}_{\Omega_{N-1}\setminus\{2\}}) = \mathrm{st}(L_k \,\mathrm{id}_{\Omega_{N-1}\setminus\{1\}}) = A_m$. The choices $y = 1, 2$ therefore give A_m, and $y = 3$ gives B_m. □

In the formulas below, $\Vert_{j=a}^{b} W_j$ denotes $W_a \,\|\, W_{a+1} \,\|\, \cdots \,\|\, W_b$ when $a \le b$, and the empty word when $a > b$.

For example, $\Vert_{j=1}^{2} [\![j, j+2]\!] = [\![1,3]\!] \,\|\, [\![2,4]\!] = [\![1,3,2,4]\!]$. The next two formulas use this concatenation convention.

Direct expansion of (4) gives

$$\begin{aligned} A_m = {} & [\![3m+1, 3m]\!] \,\|\, \Vert_{j=1}^{m} [\![3j-1, 3m+3j+1, 3m+3j]\!] \\ & \|\, [\![3m+2, 1]\!] \,\|\, \Vert_{j=m+2}^{2m} [\![3j-1, 3j-3m-2, 3j-3m-3]\!], \end{aligned} \tag{12}$$

and

$$\begin{aligned} B_m = {} & [\![3m+1, 3m, 3m+4, 3m+3]\!] \\ & \|\, \Vert_{j=2}^{m} [\![3j-1, 3m+3j+1, 3m+3j]\!] \\ & \|\, [\![3m+2, 2, 1]\!] \,\|\, \Vert_{j=m+2}^{2m} [\![3j-1, 3j-3m-2, 3j-3m-3]\!]. \end{aligned} \tag{13}$$

At $m = 1$, the two deletion words are $A_1 = [\![4,3,2,7,6,5,1]\!]$, $B_1 = [\![4,3,7,6,5,2,1]\!]$.

Paths after one-entry deletion. We construct paths of $2m+1$ moves from A_m and B_m to their identities. We use the notation $\pi \xrightarrow{a,b,c} \sigma$ when $\sigma = \mathrm{red}(s_{a,b,c}\pi)$.

For a chain $\pi_0 \xrightarrow{a_0,b_0,c_0} \pi_1 \xrightarrow{a_1,b_1,c_1} \cdots \xrightarrow{a_{r-1},b_{r-1},c_{r-1}} \pi_r$, put $n_i = \deg(\pi_i)$. At the ith step, $0 \le a_i < b_i < c_i \le n_i$, $\pi_{i+1} = \mathrm{red}(s_{a_i,b_i,c_i}\pi_i)$. Lemma 2.1 and the triangle inequality give $d_{n_i}(\pi_i, \mathrm{id}_{\Omega_{n_i}}) \le 1 + d_{n_{i+1}}(\pi_{i+1}, \mathrm{id}_{\Omega_{n_{i+1}}})$. Consequently, $d_{n_0}(\pi_0, \mathrm{id}_{\Omega_{n_0}}) \le r + d_{n_r}(\pi_r, \mathrm{id}_{\Omega_{n_r}})$; if $\pi_r = [\![1]\!]$, the right-hand side is r.

For example, $s_{0,1,2}[\![2,1,3]\!] = [\![1,2,3]\!]$ and $\mathrm{red}([\![1,2,3]\!]) = [\![1]\!]$, so $[\![2,1,3]\!] \xrightarrow{0,1,2} [\![1]\!]$ denotes one move.

The following lemma supplies the standardisation in every arrow.

Lemma 3.5. *Let* $w = [\![w_1, \dots, w_n]\!] \in \mathrm{Sym}_n$ *and put* $J = \{w_i : 1 \le i < n,\ w_{i+1} = w_i + 1\}$. *Put* $A = \Omega_n \setminus \{j+1 : j \in J\}$. *Then* $\mathrm{red}(w) = \mathrm{st}(w\,\mathrm{id}_A)$, *and each surviving value* $y \in A$ *is replaced by* $y - |\{j \in J : j < y\}|$.

If $w = s_{a,b,c}\pi$ *and* $\mathrm{red}(\pi) = \pi$, *then* $J = \{w_i : i \in \{a, a+c-b, c\} \cap \{1, \dots, n-1\},\ w_{i+1} = w_i + 1\}$.

Proof. The deleted values are exactly the noninitial values of the runs. For each $y \in A$, its rank is $|A \cap \Omega_y| = y - |\{j \in J : j+1 \le y\}| = y - |\{j \in J : j < y\}|$.

Writing $\pi = [\![P \mid U \mid V \mid Q]\!]$ gives $w = [\![P \mid V \mid U \mid Q]\!]$. The boundaries in w are after positions $a, a+c-b, c$. At every other position i, the pair $[\![w_i, w_{i+1}]\!]$ already occurs consecutively in π. Thus $\mathrm{red}(\pi) = \pi$ implies $\{i : 1 \le i < n,\ w_{i+1} = w_i + 1\} \subseteq \{a, a+c-b, c\}$. □

For the cut decompositions in this section, put $P_{m,j} = [\![3j-1, 3m+3j+1, 3m+3j]\!]$, $Q_{m,j} = [\![3j-1, 3j-3m-2, 3j-3m-3]\!]$ (14).

We start with A_m. Put $X_1 = [\![3, 2, 5, 4, 1, 6]\!]$, and, for $m \ge 2$, put

$$\begin{aligned} X_m = {} & [\![3m, 3m-1, 3m+2]\!] \\ & \Big\| \Big\|_{s=1}^{m-1} [\![3s+1, 3m+3s+2, 3m+3s+1]\!] \\ & \| [\![3m+1, 1, 3m+3]\!] \\ & \Big\| \Big\|_{s=1}^{m-2} [\![3s, 3s-1, 3m+3s+3]\!] \\ & \| [\![3m-3, 3m-4]\!]. \end{aligned} \tag{15}$$

For $m = 2$ put $Z_2 = [\![4, 3, 6, 2, 8, 7, 5, 1]\!]$, and, for $m \ge 3$, put

$$\begin{aligned} Z_m = {} & [\![3m-2, 3m-3, 3m]\!] \\ & \Big\| \Big\|_{s=1}^{m-1} [\![3s-1, 3m+3s-1, 3m+3s-2]\!] \\ & \| [\![3m-1, 1, 3m+3]\!] \\ & \Big\| \Big\|_{s=1}^{m-3} [\![3s+1, 3s, 3m+3s+3]\!] \\ & \| [\![3m-5, 3m-6]\!]. \end{aligned} \tag{16}$$

Lemma 3.6. *Let* $m \ge 1$. *Then* $A_m \xrightarrow{2,4,3m+4} X_m$. *If* $m \ge 2$, *then* $X_m \xrightarrow{3,3m+2,3m+4} Z_m \xrightarrow{3,5,3m+2} X_{m-1}$ (17). *Moreover,* $X_1 \xrightarrow{1,3,5} [\![2, 1, 3]\!] \xrightarrow{0,1,2} [\![1]\!]$.

Proof. The degrees are $\deg(A_m) = 6m+1$ for $m \ge 1$, and $\deg(X_m) = 6m-1$, $\deg(Z_m) = 6m-4$ for $m \ge 2$; also $\deg(X_1) = 6$. These degrees verify $0 \le a < b < c \le \deg(w)$ for every cut triple in the statement. For $m \ge 1$, using (14),

$$A_m = [\![3m+1, 3m \mid 2, 3m+4 \mid 3m+3, \Big\|_{j=2}^{m} P_{m,j}, 3m+2, 1 \quad \mid \Big\|_{j=m+2}^{2m} Q_{m,j}]\!].$$

The move creates the successor adjacencies $[\![1, 2]\!]$ and, when $m \ge 2$, $[\![3m+4, 3m+5]\!]$. Thus $J = \{1, 3m+4\}$ for $m \ge 2$, and $J = \{1\}$ for $m = 1$. Applying the rank formula in Lemma 3.5 to the remaining entries gives X_m in (15), with the stated word X_1 at the endpoint.

For $m \ge 3$, apply the formula for J in Lemma 3.5 with $w = s_{3,3m+2,3m+4}X_m$ or $w = s_{3,5,3m+2}Z_m$ to obtain

source	(a, b, c)	J	result after contraction
X_m	$(3, 3m+2, 3m+4)$	$\{1, 3, 3m+2\}$	Z_m
Z_m	$(3, 5, 3m+2)$	$\{1, 3m, 3m+2\}$	X_{m-1}

In the first row, contraction deletes $2, 4, 3m+3$ and replaces each remaining value by

$$y \longmapsto \begin{cases} y, & y = 1, \\ y-1, & y = 3, \\ y-2, & 5 \le y \le 3m+2, \\ y-3, & 3m+4 \le y \le 6m-1. \end{cases}$$

Applying this map to $s_{3,3m+2,3m+4}X_m$ gives the word Z_m in (16).

In the second row, contraction deletes $2, 3m+1, 3m+3$ and replaces each remaining value by

$$y \longmapsto \begin{cases} y, & y = 1, \\ y-1, & 3 \le y \le 3m, \\ y-2, & y = 3m+2, \\ y-3, & 3m+4 \le y \le 6m-4. \end{cases}$$

Applying this map to $s_{3,5,3m+2}Z_m$ gives X_{m-1} in (15).

Equations (15) and (16) give $(X_m)_{i+1} - (X_m)_i \neq 1$ for $1 \le i < 6m-1$, and $(Z_m)_{i+1} - (Z_m)_i \neq 1$ for $1 \le i < 6m-4$. Thus $\mathrm{red}(X_m) = X_m$ and $\mathrm{red}(Z_m) = Z_m$, as required in Lemma 3.5.

For $m = 2$ the two calculations are

$$\begin{aligned} X_2 &= [\![6,5,8 \mid 4,11,10,7,1 \mid 9,3 \mid 2]\!] \\ &\xrightarrow{3,8,10} [\![4,3,6 \mid 2,8 \mid 7,5,1 \mid]\!] = Z_2 \\ &\xrightarrow{3,5,8} [\![3,2,5,4,1,6]\!] = X_1. \end{aligned}$$

□

Corollary 3.7. *For $m \ge 1$, $d_{\deg(X_m)}(X_m, \mathrm{id}_{\Omega_{\deg(X_m)}}) \le 2m$ and $d_{6m+1}(A_m, \mathrm{id}_{\Omega_{6m+1}}) \le 2m+1$.*

Proof. By Lemma 3.6, there are $2(m-1)+2 = 2m$ arrows from X_m to $[\![1]\!]$. Prepending the arrow from A_m gives $2m+1$ arrows. Lemma 2.1 gives the two bounds. □

For the first arrow at $m = 1$, the move and contraction are separately $s_{2,4,7}A_1 = [\![4,3,6,5,1,2,7]\!]$, $\mathrm{red}(s_{2,4,7}A_1) = \mathrm{st}([\![4,3,6,5,1,7]\!]) = [\![3,2,5,4,1,6]\!] = X_1$. Here $J = \{1\}$, so only the entry 2 is deleted in the contraction.

For B_m we need three initial moves. For $m \ge 2$ put

$$\begin{aligned} U_m = [\![&1, 3m+4, 3, 3m, 3m-1, 3m+3, 3m+2]\!] \\ &\| \mathop{\|}_{s=1}^{m-1} [\![3s+1, 3m+3s+3, 3m+3s+2]\!] \\ &\| [\![3m+1, 2]\!] \\ &\| \mathop{\|}_{j=m+3}^{2m} [\![3j-2, 3j-3m-3, 3j-3m-4]\!]. \end{aligned} \tag{18}$$

Put $V_2 = [\![1,9,6,2,5,4,8,7,3,10]\!]$, and, for $m \ge 3$, put

$$\begin{aligned} V_m = [\![&1, 3m+3]\!] \\ &\| \mathop{\|}_{s=2}^{m-1} [\![3s, 3m+3s, 3m+3s-1]\!] \\ &\| [\![3m, 2, 3m-1, 3m-2, 3m+2, 3m+1, 3, 3m+4]\!] \\ &\| \mathop{\|}_{s=1}^{m-3} [\![3s+2, 3s+1, 3m+3s+4]\!] \\ &\| [\![3m-4, 3m-5]\!]. \end{aligned} \tag{19}$$

For $m \geq 2$ put $Y_m = \mathrm{id}_{\Omega_1} \oplus X_{m-1}$ (20).

Corollaries 2.7 and 3.7 give $d_{\deg(Y_m)}(Y_m, \mathrm{id}_{\Omega_{\deg(Y_m)}}) = d_{\deg(X_{m-1})}(X_{m-1}, \mathrm{id}_{\Omega_{\deg(X_{m-1})}}) \leq 2(m-1)$.

The following three moves reduce B_m to this family.

Lemma 3.8. *Let $m \geq 2$. Then $B_m \xrightarrow{0,3m+3,3m+6} U_m \xrightarrow{2,9,3m+6} V_m \xrightarrow{1,3m-3,3m+1} Y_m$ (21). Moreover, $B_1 = [\![4,3,7,6,5,2,1]\!] \xrightarrow{1,4,6} [\![3,2,5,4,1]\!] \xrightarrow{1,3,5} [\![2,1,3]\!] \xrightarrow{0,1,2} [\![1]\!]$.*

Proof. Let $m \geq 2$. We have $\deg(B_m) = 6m+1$, $\deg(U_m) = 6m$ and $\deg(V_m) = 6m-3$ for $m \geq 3$, whereas $\deg(V_2) = 10$.

The three cut triples in (21) satisfy $0 < 3m+3 < 3m+6 \leq 6m+1 = \deg(B_m)$, $0 \leq 2 < 9 < 3m+6 \leq 6m = \deg(U_m)$, and $0 \leq 1 < 3m-3 < 3m+1 \leq \deg(V_m)$.

Using (14),

$$B_m = [\![\mid 3m+1, 3m, 3m+4, 3m+3, \mathop{\Vert}_{j=2}^{m} P_{m,j}, 3m+2, 2$$
$$\mid 1, 3m+5, 4 \mid 3, \mathop{\Vert}_{j=m+3}^{2m} Q_{m,j}]\!].$$

The move creates the successor adjacency $[\![2,3]\!]$. Lemma 3.5, with $J = \{2\}$, gives the word U_m in (18).

For the remaining moves, Lemma 3.5 gives

source	(a,b,c)	J	result	range
U_m	$(2, 9, 3m+6)$	$\{2, 3m+4, 3m+6\}$	V_m	$m \geq 3$
U_2	$(2, 9, 12)$	$\{2, 10\}$	V_2	$m = 2$
V_m	$(1, 3m-3, 3m+1)$	$\{1, 3m, 3m+2\}$	Y_m	$m \geq 3$

For $m \geq 3$, the first row deletes $3, 3m+5, 3m+7$ and uses

$$y \longmapsto \begin{cases} y, & 1 \leq y \leq 2, \\ y-1, & 4 \leq y \leq 3m+4, \\ y-2, & y = 3m+6, \\ y-3, & 3m+8 \leq y \leq 6m. \end{cases}$$

Applied to $s_{2,9,3m+6}U_m$, this gives V_m in (19).

For $m = 2$, delete $3, 11$ from $s_{2,9,12}U_2$ and apply

$$y \longmapsto \begin{cases} y, & 1 \leq y \leq 2, \\ y-1, & 4 \leq y \leq 10, \\ y-2, & y = 12. \end{cases}$$

The result is $V_2 = [\![1,9,6,2,5,4,8,7,3,10]\!]$. In the last row, substitution of the rank formula in (19) gives $\mathrm{id}_{\Omega_1} \oplus X_{m-1} = Y_m$.

The displayed formulas give $\mathrm{red}(U_m) = U_m$ and $\mathrm{red}(V_m) = V_m$ for $m \geq 2$; thus the formula for J in Lemma 3.5 applies to each row. At the remaining endpoint, $V_2 \xrightarrow{1,3,7} [\![1,4,3,6,5,2,7]\!] = Y_2$. This proves every arrow in (21).

□

Corollary 3.9. *For every $m \geq 1$, $d_{6m+1}(B_m, \mathrm{id}_{\Omega_{6m+1}}) \leq 2m+1$.*

Proof. For $m \geq 2$, Lemmas 2.1 and 3.8, with Corollaries 2.7 and 3.7, give

$$\begin{aligned} d_{6m+1}(B_m, \mathrm{id}_{\Omega_{6m+1}}) &\leq 3 + d_{\deg(Y_m)}(Y_m, \mathrm{id}_{\Omega_{\deg(Y_m)}}) \\ &= 3 + d_{\deg(X_{m-1})}(X_{m-1}, \mathrm{id}_{\Omega_{\deg(X_{m-1})}}) \\ &\leq 3 + 2(m-1) = 2m+1. \end{aligned}$$

For $m = 1$, the displayed three-arrow chain gives $d_7(B_1, \mathrm{id}_{\Omega_7}) \leq 3$. □

For example, $Y_2 = \mathrm{id}_{\Omega_1} \oplus X_1 = [\![1,4,3,6,5,2,7]\!]$. The two remaining moves are $Y_2 \xrightarrow{2,4,6} [\![1,3,2,4]\!] \xrightarrow{1,2,3} [\![1]\!]$. They are the two moves for X_1 with all cuts increased by one, followed by the indicated run contractions. Together with the three initial steps, they give a path of length five from B_2 to the identity.

Theorem 3.10. *For odd $k \geq 3$, $d_{3k-1}(L_k, \mathrm{id}_{\Omega_{3k-1}}) = k+1$, and every standardised one-entry deletion has distance k.*

Proof. Write $k = 2m+1$ and let σ be any standardised one-entry deletion of L_k. Lemmas 3.2, 2.4, 3.4, together with Corollaries 3.7 and 3.9, give

$$\begin{aligned} k+1 &\leq d_{6m+2}(L_k, \mathrm{id}_{\Omega_{6m+2}}) \leq d_{6m+1}(\sigma, \mathrm{id}_{\Omega_{6m+1}}) + 1 \\ &\leq 1 + \max\{d_{6m+1}(A_m, \mathrm{id}_{\Omega_{6m+1}}), d_{6m+1}(B_m, \mathrm{id}_{\Omega_{6m+1}})\} \\ &\leq 2m+2 = k+1. \end{aligned}$$

Every inequality is therefore an equality. □

3.2. **The even case.** Let $k = 2m+2 \geq 4$, $n = 3(k-1)-1 = 6m+2$, and $L = L_{k-1}$. Put $u = 3m-2$ and $v = 6m$. By (11), $(u\,v\,n)$ is a factor of κ_L.

The definition of L gives $L_n = u$, $nL^{-1} < vL^{-1} < uL^{-1} = n$. Thus, starting at u in w_L, one encounters n before v. Write $L = P \parallel [\![n]\!] \parallel Q \parallel [\![u]\!]$, $E_k = L \oplus [\![2,1]\!] = P \parallel [\![n]\!] \parallel Q \parallel [\![u, n+2, n+1]\!]$. This is the permutation defined in (5).

Lemma 3.11. *Let $k \geq 4$ be even. The commutator of E_k is obtained from $\kappa_{L_{k-1}}$ by replacing the factor $(u\,v\,n)$ with $(u,\, v,\, n,\, n+2,\, n+1)$. The five labels of this 5-cycle are encountered in the order $u, n+2, n+1, n, v$ when w_{E_k} is traversed starting at u.*

Proof. Equation (7) gives

$$x\kappa_{E_k} = \begin{cases} x\kappa_L, & 0 \leq x < n, \\ n+2, & x = n, \\ n+1, & x = n+2, \\ u, & x = n+1. \end{cases}$$

Together with $u\kappa_L = v$ and $v\kappa_L = n$, this gives the cycle $u \longmapsto v \longmapsto n \longmapsto n+2 \longmapsto n+1 \longmapsto u$. All other cycle factors are unchanged. Since u is the last entry of L and u, n, v occur in that cyclic order, there are words A, B, C such that $w_L = (u\,A\,n\,B\,v\,C)$, $w_{E_k} = (u\,n+2\,n+1\,A\,n\,B\,v\,C)$. This gives the required order of the five labels. □

Corollary 3.12. *For even $k \geq 4$, the cycle type of κ_{E_k} is $5\,3^{k-2}$.*

Proof. Lemma 3.1 gives $k-1$ disjoint 3-cycles for $\kappa_{L_{k-1}}$. Lemma 3.11 replaces one of them by a 5-cycle. □

The factor for $k = 4$ is shown in Figure 3. The solid arrows give its orientation. Together with two consecutive solid arrows, the dashed closing chords give $\tau_1 = (8\,10\,9)$ and $\tau_2 = (10\,9\,1)$, the only 3-cycles τ such that $\tau w_{E_4} \in \mathcal{C}_{11}$ and $c_+(\kappa_{E_4}\tau^{-1}) = c_+(\kappa_{E_4}) + 2$.

Put $p = nE_k^{-1}$. The generators $s \in S_{n+2}$ with $\nu(sE_k) = \nu(E_k) - 1$ are $s_{p,n+1,n+2}$ and $s_{n,n+1,n+2}$. For either generator, $\mathrm{red}(sE_k) = L_{k-1} \oplus \mathrm{id}_{\Omega_1}$.

Lemma 3.13. *Let $k \geq 4$ be even. Then $d_{3k-2}(E_k, \mathrm{id}_{\Omega_{3k-2}}) = k+1$.*

Proof. The commutator has $k-1$ even cycle factors on $3k-1$ labels. Therefore $\nu(E_k) = \frac{(3k-1)-(k-1)}{2} = k$.

For each 3-cycle factor τ inherited from κ_L, $\tau w_{E_k} \notin \mathcal{C}_{n+3}$, because the order of its three labels in w_{E_k} is the same as in w_L. By Lemma 2.13, a step with $\nu(sE_k) = \nu(E_k) - 1$ must

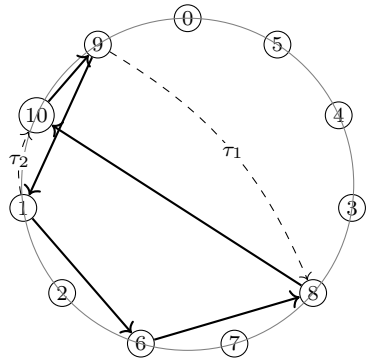


FIGURE 3. The cycle factor $(1\,6\,8\,10\,9)$ of κ_{E_4}.

therefore correspond to $\tau \in \{(u\,v\,n), (v,\, n,\, n+2), (n,\, n+2,\, n+1), (n+2,\, n+1,\, u), (n+1,\, u,\, v)\}$.

The five candidates and their orders in w_{E_k} are

τ	order in w_{E_k}	$\tau w_{E_k} \in \mathcal{C}_{n+3}$
$(u\,v\,n)$	u, n, v	no
$(v,\, n,\, n+2)$	$v, n+2, n$	no
$(n,\, n+2,\, n+1)$	$n, n+2, n+1$	yes
$(n+2,\, n+1,\, u)$	$n+2, n+1, u$	yes
$(n+1,\, u,\, v)$	$n+1, v, u$	no

by Lemmas 3.11 and 2.8. Thus precisely $\tau_1 = (n,\, n+2,\, n+1)$ and $\tau_2 = (n+2,\, n+1,\, u)$ give decreasing moves.

Let p be the position of n in E_k. By Lemma 2.10, the corresponding moves are $s_{p,n+1,n+2}[\![P, n \mid Q, u, n+2 \mid n+1 \mid]\!] = [\![P, n \mid n+1 \mid Q, u, n+2 \mid]\!]$, $s_{n,n+1,n+2}[\![P, n, Q, u \mid n+2 \mid n+1 \mid]\!] = [\![P, n, Q, u \mid n+1 \mid n+2 \mid]\!]$.

Put $A = \Omega_{n+2} \setminus \{n+1\}$. For either generator s, $\mathrm{st}((sE_k)\,\mathrm{id}_A) = \mathrm{st}([\![P, n, Q, u, n+2]\!]) = L_{k-1} \oplus \mathrm{id}_{\Omega_1}$. The omitted entry $n+1$ belongs to the run containing n in the first word and $n+2$ in the second. Thus A meets every run in both words, and by Lemma 2.1 preserves both distances.

By Theorem 3.10 and Lemma 2.6, $d_{n+1}(L_{k-1} \oplus \mathrm{id}_{\Omega_1}, \mathrm{id}_{\Omega_{n+1}}) = d_n(L_{k-1}, \mathrm{id}_{\Omega_n}) = k$. For each of these two generators s, Lemma 2.1 gives $d_{n+2}(sE_k, \mathrm{id}_{\Omega_{n+2}}) = k$. Lemma 2.14 therefore implies $d_{n+2}(E_k, \mathrm{id}_{\Omega_{n+2}}) \geq k+1$.

Conversely, Lemma 2.6 gives $d_{n+2}(E_k, \mathrm{id}_{\Omega_{n+2}}) \leq d_n(L_{k-1}, \mathrm{id}_{\Omega_n}) + d_2([\![2,1]\!], \mathrm{id}_{\Omega_2}) = k+1$. Since $n+2 = 3k-2$, this proves the assertion. □

We finish the proof of Theorem 1.2. For $y \in \{n+1, n+2\}$, Lemma 2.6 gives $\mathrm{st}(E_k\,\mathrm{id}_{\Omega_{n+2}\setminus\{y\}}) = L \oplus \mathrm{id}_{\Omega_1}$, since either one-entry deletion of $[\![2,1]\!]$ standardises to id_{Ω_1}.

By Theorem 3.10 and Corollary 2.7, $d_{n+1}(L \oplus \mathrm{id}_{\Omega_1}, \mathrm{id}_{\Omega_{n+1}}) = d_n(L, \mathrm{id}_{\Omega_n}) = k$. For $x \in \Omega_n$, Lemma 2.6 gives $\mathrm{st}(E_k\,\mathrm{id}_{\Omega_{n+2}\setminus\{x\}}) = \mathrm{st}(L\,\mathrm{id}_{\Omega_n\setminus\{x\}}) \oplus [\![2,1]\!]$.

By Theorem 3.10 and Lemma 2.6,

$$\begin{aligned} d_{n+1}(\mathrm{st}(E_k\,\mathrm{id}_{\Omega_{n+2}\setminus\{x\}}), \mathrm{id}_{\Omega_{n+1}}) &\leq d_{n-1}(\mathrm{st}(L\,\mathrm{id}_{\Omega_n\setminus\{x\}}), \mathrm{id}_{\Omega_{n-1}}) + d_2([\![2,1]\!], \mathrm{id}_{\Omega_2}) \\ &= (k-1)+1 = k. \end{aligned}$$

Lemma 3.13 gives $d_{n+2}(E_k, \mathrm{id}_{\Omega_{n+2}}) = k+1$;

For every standardised one-entry deletion σ of E_k, Lemma 2.4 and the bounds just proved give $k = d_{n+2}(E_k, \mathrm{id}_{\Omega_{n+2}}) - 1 \leq d_{n+1}(\sigma, \mathrm{id}_{\Omega_{n+1}}) \leq k$. Together with Theorem 3.10, this proves Theorem 1.2.

It remains to determine which values occur in

$$3k - b_k \in \begin{cases} \{-1, 0, 1\}, & k \geq 3 \text{ odd}, \\ \{-1, 0, 1, 2\}, & k \geq 4 \text{ even}. \end{cases}$$

In particular, is $3k - b_k = 1$ for every odd $k \geq 3$? For even $k \geq 4$, is $b_k = 3k - 2$, or does a minimal excluded permutation exist in some Sym_n with $n \geq 3k - 1$?

4. Genome rearrangements and previous results

A model of genome rearrangement records the order of distinct markers in a linear chromosome. Label the common markers $1, \ldots, n$ in their order in a reference chromosome. Their order in a second chromosome is then the word of a permutation $\pi \in \mathrm{Sym}_n$; see Figure 4. A *block transposition* exchanges two adjacent non-empty segments, preserving the order inside each segment. By Equation (2), the exchange with boundaries a, b, c is exactly $\pi \mapsto s_{a,b,c}\pi$. Equal labels in Figure 4 denote the same marker, so the same eight markers appear in two orders.

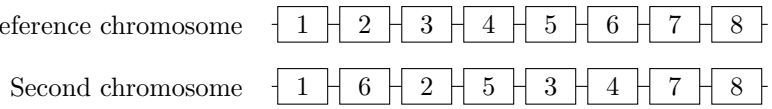


Figure 4. Two orders of the same eight markers.

In the terminology of the genome-rearrangement literature, sorting a permutation means transforming its word to the reference order $[\![1, \ldots, n]\!]$. A sequence of r block transpositions gives $s_r \cdots s_1 \pi = \mathrm{id}_{\Omega_n}$, equivalently $\pi = s_1^{-1} \cdots s_r^{-1}$ with all factors in S_n. Thus its transposition distance is exactly $d_n(\pi, \mathrm{id}_{\Omega_n})$. Retaining a set A of marker values gives the partial bijection $\pi\,\mathrm{id}_A$. The permutation $\mathrm{st}(\pi\,\mathrm{id}_A)$ is the corresponding standardised subword, called a *pattern*.

For $k \geq 1$, put $\mathcal{P}_k = \bigcup_{n \geq 0} \{\pi \in \mathrm{Sym}_n : d_n(\pi, \mathrm{id}_{\Omega_n}) \leq k\}$. Write $\sigma \preccurlyeq \pi$ when σ is a standardised restriction of π, and $\sigma \prec \pi$ when the restriction is proper. Equation (3) states that $\pi \in \mathcal{P}_k$ and $\sigma \preccurlyeq \pi$ imply $\sigma \in \mathcal{P}_k$. Thus $\mathcal{P}_k$ is a *permutation class*, meaning a set closed under standardised restrictions. Its *forbidden-pattern basis* is $\mathcal{B}_k = \{\beta \notin \mathcal{P}_k : \sigma \in \mathcal{P}_k \text{ for every } \sigma \prec \beta\}$. These are exactly the permutations minimal excluded at radius k. Every permutation outside $\mathcal{P}_k$ contains a member of $\mathcal{B}_k$: choose a restriction of smallest degree that still lies outside $\mathcal{P}_k$. Therefore $\pi \in \mathcal{P}_k$ if and only if no $\beta \in \mathcal{B}_k$ satisfies $\beta \preccurlyeq \pi$.

The bound of Cerbai and Ferrari [2, Theorem 2.5] gives $\mathcal{B}_k \subseteq \bigcup_{j=0}^{3k+1} \mathrm{Sym}_j$. In particular, this basis is finite and $b_k = \max\{\deg(\beta) : \beta \in \mathcal{B}_k\}$. The contribution of Theorem 1.2 is to construct members of these bases in degrees close to this upper bound: $L_k \in \mathcal{B}_k \cap \mathrm{Sym}_{3k-1}$ for odd $k \geq 3$, and $E_k \in \mathcal{B}_k \cap \mathrm{Sym}_{3k-2}$ for even $k \geq 4$. Hence the upper bound exceeds the largest basis degree by at most 2 for odd $k \geq 3$ and at most 3 for even $k \geq 4$. The leading coefficient 3 is optimal: no bound $b_k \leq ck + O(1)$ with $c < 3$ can hold as $k \to \infty$. These bounds leave the exact value of $3k - b_k$ and the description of $\mathcal{B}_k$ for general k open.

There is also an equivalent statement about detecting distance greater than k by examining restrictions. For every $\pi \in \mathrm{Sym}_n$, $d_n(\pi, \mathrm{id}_{\Omega_n}) > k \iff \exists A \subseteq \Omega_n : |A| \leq b_k$ and $d_{|A|}(\mathrm{st}(\pi\,\mathrm{id}_A), \mathrm{id}_{\Omega_{|A|}}) > k$. The forward implication follows by choosing a contained basis element; the reverse implication is (3). Moreover, b_k is the smallest possible uniform cutoff in this equivalence. A basis element of degree b_k has distance greater than k, but every restriction of smaller degree has distance at most k. Our constructions therefore show that such a restriction may need at least $3k - 1$ entries for odd $k \geq 3$, and at least $3k - 2$ for even $k \geq 4$, whereas $3k + 1$ entries always suffice. Thus the optimal number of entries is $3k + O(1)$.

Bafna and Pevzner studied this distance as a model of changes in gene order [1, §1]; their journal article records its preliminary 1995 version on page 224. The Cayley-graph formulation and the inverse-closed generating set are studied by Korchmaros and Kovács [4]. Cerbai and Ferrari prove the contraction property and the bound $3k + 1$ for basis elements [2, §2 and Theorem 2.5].

The cycle called $\overline{\pi}$ in the algebraic formulation of [5, pp. 3–4] is w_π here. Its applicable 3-cycles are precisely those τ for which $\tau w_\pi \in \mathcal{C}_{n+1}$. With right actions, the product of cycles used there is $w_\pi^{-1}\rho_{n+1} = w_\pi^{-1}\kappa_\pi w_\pi$. It is therefore conjugate to our commutator. Their count of cycles of odd length is c_+, since a cycle of odd length is an even permutation. The lower bound and cycle-count change formula used here are recorded there as Lemma 10 and Proposition 12, respectively.

Silva, Kowada and Walter construct permutations of degree $6r - 1$ and exact distance $\lceil 11r/4 \rceil$ [6, Theorem 1]. Their Example 6 already contains L_3 and a four-step sorting sequence. Theorem 1.2 imposes the additional condition that every one-entry deletion lowers the distance by exactly one. It gives basis elements of degrees $3k - 1$ and $3k - 2$ according to the parity of k, and hence determines the leading coefficient of the maximum basis degree.

In the notation of this paper, their result is $P_r \in \mathrm{Sym}_{6r-1}$, $d_{6r-1}(P_r, \mathrm{id}_{\Omega_{6r-1}}) = \left\lceil \frac{11r}{4} \right\rceil$ $(r \geq 1)$. On the other hand, for every $k \geq 3$, we find

$$(\pi, n) = \begin{cases} (L_k, 3k-1), & k \text{ odd}, \quad d_n(\pi, \mathrm{id}_{\Omega_n}) = k+1, \\ (E_k, 3k-2), & k \text{ even}, \quad d_{n-1}(\sigma, \mathrm{id}_{\Omega_{n-1}}) = k \end{cases}$$

for every standardised one-entry deletion σ of π. To compare examples of the same distance, put $k = \lceil 11r/4 \rceil - 1$. For $r \geq 2$, $d_n(\pi, \mathrm{id}_{\Omega_n}) = d_{6r-1}(P_r, \mathrm{id}_{\Omega_{6r-1}}) = k + 1$, $n \geq 3k - 2 = 3\left\lceil \frac{11r}{4} \right\rceil - 5 > 6r - 1$. Thus our examples have larger degree at the same distance. The deletion equality also proves $b_k \geq n$. Their distance formula alone does not establish $b_{\lceil 11r/4 \rceil - 1} \geq 6r - 1$, since it supplies no deletion condition. For distance divided by degree, their examples give the larger limit: $\lim_{r\to\infty} \frac{d_{6r-1}(P_r, \mathrm{id}_{\Omega_{6r-1}})}{6r-1} = \frac{11}{24} > \frac{1}{3}$ $= \lim_{k\to\infty} \frac{d_n(\pi, \mathrm{id}_{\Omega_n})}{n}$. The results therefore concern different bounds: theirs concerns sorting distance, whereas ours gives $b_k \geq 3k - 2$ and $b_k/k \to 3$.

In the genome model, a minimal excluded permutation is a minimal marker-order witness excluding every history of at most k block transpositions. For example, the orders $[\![1, 2, 3]\!]$ and $[\![3, 2, 1]\!]$ require two exchanges, whereas omitting any one marker leaves two orders connected by one exchange. Theorem 1.2 gives corresponding examples requiring $k + 1$ exchanges, with every one-marker omission requiring exactly k. The maximum number of markers in such a minimal exclusion is asymptotic to $3k$. Here omission means removing a marker from the comparison. The distance measures the fewest events compatible with the recorded orders; it does not determine their historical number or probability.

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